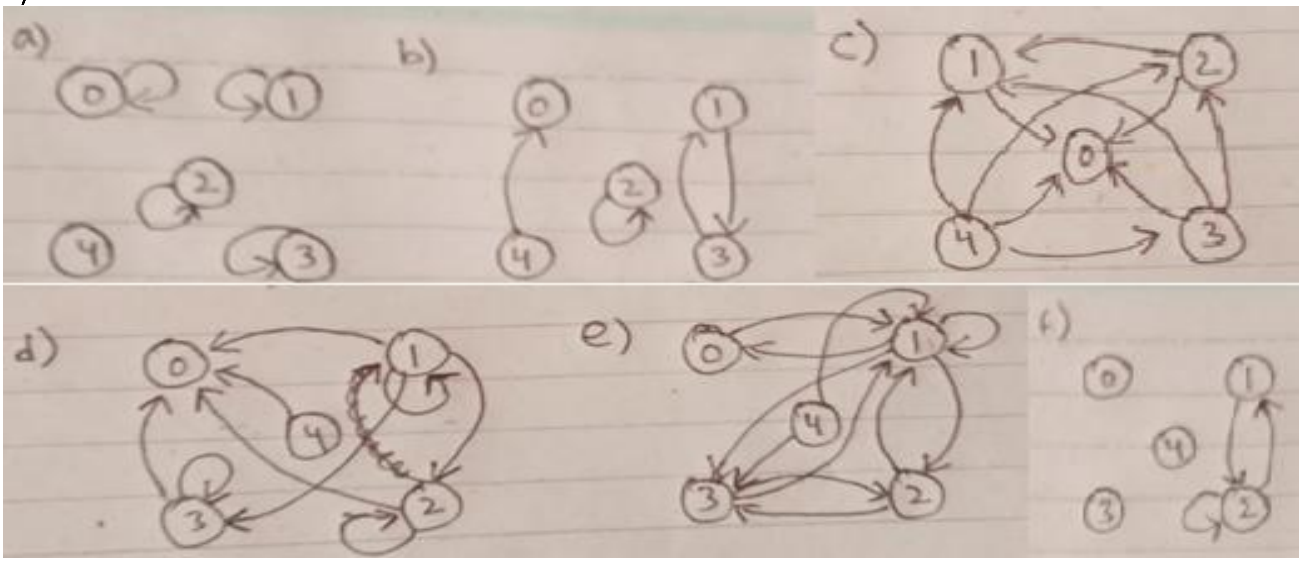
CS1005-Discrete Structures Fall 2025
Assignment # 2 -- Solution

1. **i) Solution:**
   - a) {(0,0), (1, 1), (2, 2), (3, 3)}
   - b) {(1, 3), (2, 2), (3, 1), (4, 0)}
   - c) {(1, 0), (2, 0), (3, 0), (4, 0), (2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)}
   - d) {(1, 0), (2, 0), (3, 0), (4, 0), (1, 1), (1,2), (2,2), (1,3), (3,3)}
   - e) {(1,0), (0,1), (1,1), (1,2), (1,3), (2,1), (3,1), (4,1), (2,3), (3,2), (4,3)}
   - f) {(1,2), (2,1), (2,2)}

   ii) Solution:

$$a)\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\\0&0&0&0\end{bmatrix}\quad b)\begin{bmatrix}0&0&0&0\\0&0&0&1\\0&0&1&0\\0&1&0&0\\1&0&0&0\end{bmatrix}\quad c)\begin{bmatrix}0&0&0&0\\1&0&0&0\\1&1&0&0\\1&1&1&0\\1&1&1&1\end{bmatrix}\quad d)\begin{bmatrix}0&0&0&0\\1&1&1&1\\1&0&1&0\\1&0&0&1\\1&0&0&0\end{bmatrix}\quad e)\begin{bmatrix}0&1&0&0\\1&1&1&1\\0&1&0&1\\0&1&1&1\\0&1&0&1\end{bmatrix}\quad f)\begin{bmatrix}0&0&0&0\\0&0&1&0\\0&1&1&0\\0&0&0&0\\0&0&0&0\end{bmatrix}$$

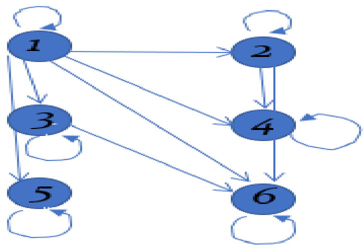
   iii) Solution:



   iv) Solution:
   - a) {(0,0), (1, 1), (2, 2), (3, 3)}
   - b) {(1, 3), (2, 2), (3, 1), (0, 4)}
   - c) {(0, 1), (0, 2), (0, 3), (0, 4), (1, 2), (1, 3), (2, 3), (1, 4), (2, 4), (3, 4)}
   - d) {(0, 1), (0, 2), (0, 3), (0, 4), (1, 1), (2,1), (2,2), (3,1), (3,3)}
   - e) {(1,0), (0,1), (1,1), (1,2), (1,3), (2,1), (3,1), (1,4), (2,3), (3,2), (3,4)}
   - f) {(1,2), (2,1), (2,2)}

2. **i) Solution: R= {(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)}**

$$\begin{bmatrix}1&1&1&1&1&1\\0&1&0&1&0&1\\0&0&1&0&0&1\\0&0&0&1&0&0\\0&0&0&0&1&0\\0&0&0&0&0&1\end{bmatrix}$$



   The relation is not an equivalence relation since it does not hold symmetric property.
   The relation is partial order relation since it holds Reflexive, Antisymmetric and Transitive property.

   ii) Solution:

   a) {(1,1), (2,2), (3,3), (4,4)} b) {(1,2), (2,1), (3,4)}
   c) {(1,2), (2,1), (1,1)} d) {(1,2), (2,1), (3,4)}

3. i) Solution:

(a) R is reflexive because R contains (a, a), (b, b), (c, c), and (d, d).
(b) R is not symmetric because R contains (a, c) but not (c, a) $\in$ R .
(c) R is not antisymmetric because both (b, c) $\in$ R and (c, b) $\in$ R , but b = c.
(d) R is not Transitive because both (a, c) $\in$ R and (c, b) $\in$ R , but not (a, b) $\in$ R.
(e) R is not irreflexive because R contains (a, a), (b, b), (c, c), and (d, d).
(f) R is not Asymmetric because R is not Antisymmetric.

ii)

a) Solution:
(a) R is not reflexive: It doesn't contain (1,1) and (4,4).
(b) R is not symmetric because R contains (2, 4) but not (4, 2) $\in$ R.
(c) R is not antisymmetric: we have (2,3) and (3,2) but 2 $\neq$ 3.
(d) R is Transitive because for any numbers a, b, and c, if (a, b), (b, c) $\in$ R then (a, c) $\in$ R.

b) Solution:
(a) R is reflexive: It contains (1,1), (2,2), (3,3) and (4,4).
(b) R is symmetric because (a,b) and (b,a) $\in$ R.
(c) R is not antisymmetric: we have (1,2) and (2,1) but 1 $\neq$ 2.
(d) R is Transitive because for any numbers a, b, and c, if (a, b), (b, c) $\in$ R then (a, c) $\in$ R.

c) Solution:
(a) R is not reflexive: It doesn't contain (1,1), (2,2), (3,3) and (4,4).
(b) R is symmetric because R contains (2, 4) and (4, 2) $\in$ R.
(c) R is not antisymmetric: we have (2,4) and (4,2) but 2 $\neq$ 4.
(d) R is not Transitive because (2,4), (4, 2) $\in$ R but not (2,2) $\in$ R.

d) Solution:
(a) R is not reflexive: It doesn't contain (1,1), (2,2), (3,3) and (4,4).
(b) R is not symmetric because (1,2) $\in$ R but not (2,1) $\in$ R.
(c) R is antisymmetric: we have (a,b) but not (b,a) $\in$ R.
(d) R is not Transitive because (1,2), (2, 3) $\in$ R but not (1,3) $\in$ R.

e) Solution:
(a) R is reflexive: It contains (1,1), (2,2), (3,3) and (4,4).
(b) R is symmetric because R contains (a,b) and (b,a) $\in$ R.
(c) R is antisymmetric: we have (a,b) and (b,a) $\in$ R then a = b.
(d) R is Transitive because for any numbers a, b, and c, if (a, b), (b, c) $\in$ R then (a, c) $\in$ R.

f) Solution:
(a) R is not reflexive: It doesn't contain (1,1) ,(2,2), (3,3) and (4,4).
(b) R is not symmetric because (1,4) $\in$ R but not (4,1) $\in$ R.
(c) R is not antisymmetric: we have (1,3) and (3,1) $\in$ R but 1 $\neq$ 3.
(d) R is not Transitive because we have (1,3) and (3,1) $\in$ R but not (1,1) $\in$ R.

4. i) Solution:

a)

The relation R is **not reflexive**, because a person cannot be taller than himself/herself.

The relation R is **not symmetric**, because if person A is taller than person B, then person B is NOT taller than person A.

The relation R is **antisymmetric**, because $(a, b) \in R$ and $(b, a) \in R$ cannot occur at the same time (as one person is always taller than the other, but not the other way around).

The relation R is **transitive**, because if person A is taller than person B and if person B is taller than person C, then person A needs to be taller than person C as well.

b) Solution:

The relation R is **reflexive**, because a person is born on the same day as himself/herself.

The relation R is **symmetric**, because if person A and person B are born on the same day, then person B is also born on the same day as person A.

The relation R is **not antisymmetric**, because if person A and person B are born on the same day and if person B and person A are born on the same day, then these two people are not necessarily the same person.

The relation R is **transitive**, because if person A and person B are born on the same day and if person B and person C are born on the same day, then person A and person C are also born on the same day.

c) Solution:

The relation R is **reflexive**, because a person has the same first name as himself/herself.

The relation R is **symmetric**, because if person A has the same first name as person B, then person B also has the same first name as person A.

The relation R is **not antisymmetric**, because if person A has the same first name as person B and if person B also has the same first name as person A, then these two people are not necessarily the same person (as there are different people with the same first name).

The relation R is **transitive**, because if person A has the same first name as person B and if person B also has the same first name as person C, then person A also has the same first name as person C.

d) Solution:

The relation R is **reflexive**, because a person has the same grandparents as himself/herself.

The relation R is **symmetric**, because if person A and person B have a common grandparent, then person B and person A also have a common grandparent.

The relation R is **not antisymmetric**, because if person A and person B have a common grandparent and if person B and person A have a common grandparent, then these two people are not necessarily the same person (as there are different people with the same grandparents).

The relation R is **not transitive**, because if person A and person B have a common grandparent and if person B and person C have a common grandparent, then person A and person C do not necessarily have a common grandparent (for example, the common grandparent of A and B can be from person B's father's side of the family, while the common grandparent of B and C can be from person B's mother's side of the family).

ii) **Solution:**

a) Reflexive: everyone who has visited web page *a* has also visited web page *a*.
NOT Symmetric: if everyone who has visited web page *a* has visited web page *b*, this does not mean that everyone who has visited web page *b* has also visited web page *a*.
Transitive: if everyone who has visited web page *a* also has visited web page *b* and everyone who has visited web page *b* also has visited web page *c*, then it follows that everyone who has visited web page *a* has also visited web page *c*.

b) NOT Reflexive: (a,b) ∈ R, if there exist a web-page (say web-page m) such that m has links for both web-page a and web-page b . Now, (a, a) need not belong to R, as there could be no web-page pointing to web-page a.
Symmetric: if (a, b) ∈ R, i.e there is a web-page linking to web-page a and web-page b, then (b,a) ∈ R. Hence, Symmetric.
NOT Transitive: If (a, b) ∈ R, and (b, c) ∈R then it is not necessary that (a,c) ∈ R. If web-page e points to web-page a and web-page b, web-page f points to web-page b and web-page c then it is not necessary that there exit a web-page which'll point to both web-page and web-page c.

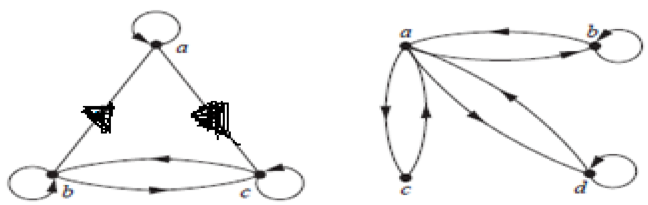
5. i) **Solution:**

R1= { (2,1), (3,1), (3,2) }
R2= { (1,1), (2,2), (3,3), (2,1), (3,1), (3,2) }
R3= { (1,2), (1,3), (2,3) }
R4= { (1,1), (2,2), (3,3), (1,2), (1,3), (2,3) }
R5= { (1,1), (2,2), (3,3) }
R6= {(1,2), (1,3), (2,1), (2,3), (3,1), (3,2) }

a) R2 ∪ R4 = { (1,1), (2,2), (3,3), (2,1), (3,1), (3,2) (1,2), (1,3), (2,3) }
b) R3 ∪ R6 = {(1,2), (1,3), (2,1), (2,3), (3,1), (3,2) }
c) R3 ∩ R6 = { (1,2), (1,3), (2,3) }
d) R4 ∩ R6 = { (1,2), (1,3), (2,3) }
e) R3 − R6 = { } OR Φ
f) R6 − R3 = { (2,1), (3,1), (3,2) }
g) R2 ⊕ R6 = { (1,1), (2,2), (3,3), (1,2), (1,3), (2,3) }
h) R3 ⊕ R5 = { (1,1), (2,2), (3,3), (1,2), (1,3), (2,3) }
i) R2 ◦ R1 = { (2,1), (3,1), (3,2) }
j) R6 ◦ R6 = { (1,1), (2,2), (3,3), (2,1), (3,1), (3,2) (1,2), (1,3), (2,3)}

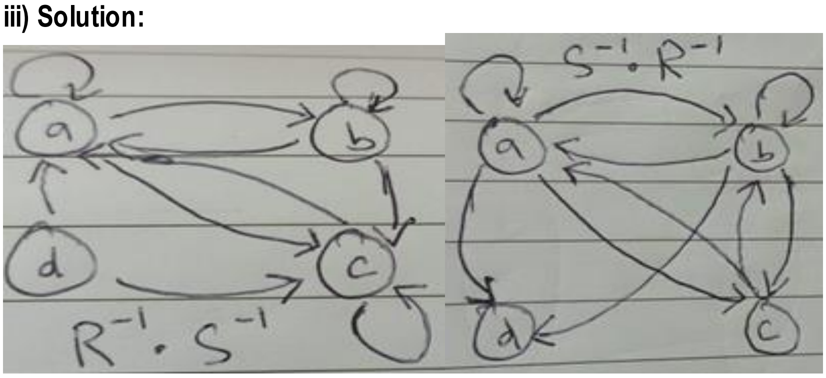
ii) **Solution:**
The pair (1. 1) is in R1, R3, R4, and R6. The pair (1, 2) is in R1 and R6. The Pair (2, 1) is in R2, R5, and R6. The pair (1, −1) is in R2, R3, and R6. The pair (2, 2) is in R1, R3, and R4.

6. i) **Solution:**
a) Because there are loops at every vertex of the directed graph of R, it is reflexive. R is neither symmetric nor antisymmetric because there is an edge from a to b but not one from b to a, but there are edges in both directions connecting b and c. Finally, R is not transitive because there is an edge from a to b and an edge from b to c, but no edge from a to c.
b) Because loops are not present at all the vertices of the directed graph of S, this relation is not reflexive. It is symmetric and not antisymmetric, because every edge between distinct vertices is accompanied by an edge in the opposite direction. It is also not hard to see from the directed graph that S is not transitive, because (c, a) and (a, b) belong to S, but (c, b) does not belong to S.
ii) **Solution:**



(a) Directed graph of R^{-1} (b) Directed graph of S^{-1}

iii) **Solution:**



7. a) Solution:
Show that all of the properties of an equivalence relation hold.
Reflexivity: Because l(a) = l(a), it follows that aRa for all strings a.
Symmetry: Suppose that aRb. Since l(a) = l(b), l(b) = l(a) also holds and bRa.
Transitivity: Suppose that aRb and bRc. Since l(a) = l(b), and l(b) = l(c), l(a) = l(a) also holds an d aRc.

b) Solution:
Recall that $a \equiv b \pmod{m}$ if and only if m divides $a - b$.
Reflexivity: $a \equiv a \pmod{m}$ since $a - a = 0$ is divisible by m since $0 = 0 \cdot m$.
Symmetry: Suppose that $a \equiv b \pmod{m}$. Then $a - b$ is divisible by m, and so $a - b = km$, where k is an integer. It follows that $b - a = (-k)\,m$, so $b \equiv a \pmod{m}$.
Transitivity: Suppose that $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$. Then m divides both $a - b$ and $b - c$. Hence, there are integers k and l with $a - b = km$ and $b - c = lm$. We obtain by adding the equations:
$a - c = (a - b) + (b - c) = km + lm = (k + l)\,m$. Therefore, $a \equiv c \pmod{m}$.

c) Solution:
R = {(p, q): |p - q| is even}. Where p, q belongs to P.
Reflexive Property:
From the provided relation |p – p| = | 0 |=0.
- And 0 is always even.
- Therefore, |p – p| is even.
- Hence, (p, p) relates to R so, R is Reflexive.

Symmetric Property:
From the given relation |p – q| = |q – p|.
- We know that |p – q| = | – (q – p)|= |q – p|
- Hence |p – q| is even.
- Next |q – p| is also even. Accordingly, if (p, q) ∈ R, then (q, p) also belongs to R.

Therefore, R is symmetric.
Transitive Property:
- If |p – q| is even, then (p-q) is even.
- Similarly, if |q-r| is even, then (q-r) is also even.
- The summation of even numbers is too even.
- So, we can address it as p – q+ q-r is even.
- Next, p – r is further even.

Accordingly,
- |p – q| and |q-r| is even, then |p – r| is even.
- Consequently, if (p, q) ∈ R and (q, r) ∈ R, then (p, r) also refers to R. Therefore, R is transitive.

Hence, R is an Equivalence Relation.

d) Solution:
Reflexive: Relation R is reflexive because (5, 5), (2, 2), (3, 3) and (4, 4) ∈ R.
Symmetric: Relation R is symmetric as whenever (a, b) ∈ R, (b, a) also relates to R.
Example: $(3, 5) \in R \Longrightarrow (5, 3) \in R$.
Transitive: Relation R is transitive as whenever (a, b) and (b, c) relate to R, (a, c) also relates to R.
Example: $(3, 5) \in R$ and $(5, 3) \in R \Longrightarrow (3, 3) \in R$.
Accordingly, R is reflexive, symmetric and transitive. So, R is an Equivalence Relation.

8. a) Solution:
Consider the set A = {1, 2, 3, 4} containing four Positive integers. Find the relation for this set such as R = {(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3), (1, 1), (2, 2), (3, 3), (4, 4)}.
Reflexive: The relation is reflexive as for every a ∈ A. (a, a) ∈ R, i.e. (1, 1), (2, 2), (3, 3), (4, 4) ∈ R.
Antisymmetric: The relation is antisymmetric as whenever (a, b) and (b, a) ∈ R, we have a = b.
Transitive: The relation is transitive as when ever (a, b) and (b, c) ∈ R, we have (a, c) ∈ R.
Example: (4, 2) ∈ R and (2, 1) ∈ R, implies (4, 1) ∈ R.
As the relation is reflexive, antisymmetric and transitive. Hence, it is a partial order relation.

b) Solution:
Reflexive: We have a divides a, ∀ a∈ N. Therefore, relation 'Divides' is reflexive.
Antisymmetric: Let a, b, c ∈N, such that a divides b. It implies b divides a iff a = b. So, the relation is antisymmetric.
Transitive: Let a, b, c ∈N, such that a divides b and b divides c.
Then a divides c. Hence the relation is transitive.
Thus, the relation being reflexive, antisymmetric and transitive, the relation 'divides' is a partial order relation.

c) Solution:
Reflexive: Relation R is reflexive because (5, 5), (2, 2), (3, 3) and (4, 4) ∈ R.
AntiSymmetric: Relation R is symmetric as whenever (a, b) ∈ R, and (b, a) ∈ R then a = b.
Example: (3, 5) ∈ R and (5, 3) ∈ R ⟹ 5 ≠ 3.
Transitive: Relation R is transitive as whenever (a, b) and (b, c) relate to R, (a, c) also relates to R.
Example: (3, 5) ∈ R and (5, 3) ∈ R ⟹ (3, 3) ∈ R.
Accordingly, R is reflexive, transitive but not antisymmetric. So, R is not a partial order relation.

d) Solution:
Show that all of the properties of partial order relation hold.
Reflexivity: Because l(a) = l(a), it follows that aRa for all strings a.
Antisymmetry: Suppose that aRb. Since l(a) = l(b), l(b) = l(a) also holds and bRa.
Transitivity: Suppose that aRb and bRc. Since l(a) = l(b), and l(b) = l(c), l(a) = l(a) also holds and aRc.

9. a)
(i) Solution: 1, 3, 7,15, 31 are the first five terms of the given sequence.
(ii) Solution: $\frac{17}{2}, 7, \frac{11}{2}, 4, \frac{5}{2}$ are the first five terms.
(iii) Solution: $-1, \frac{1}{4}, -\frac{1}{9}, \frac{1}{16}, -\frac{1}{25}$ are the first five terms.
(iv) Solution: $7, \frac{10}{3}, \frac{13}{5}, \frac{16}{7}, \frac{19}{9}$ are the first five terms.
b)
(i) Solution:
Here common difference (d) = -7
$T_n = a + (n-1)d$; $T_{11} = -15 + (11-1)(-7) = -85$

(ii) Solution:
Here common difference (d) = 3b
$T_n = a + (n-1)d$; $T_{15} = a - 42b + (15-1)(3b) = a$

(iii) Solution:
Here common ratio (r) = $\frac{3}{4}$
$T_n = ar^{n-1}$; $T_{17} = 4(\frac{3}{4})^{17-1} = \frac{3^{16}}{4^{15}}$

(iv) Solution:
Here common ratio (r) = $\frac{1}{2}$
$T_n = ar^{n-1}$; $T_{17} = 32(\frac{1}{2})^{9-1} = \frac{1}{8}$

10. a)
(i) Solution:
Since Tn = ar^(n-1)
T3=ar^2=10 ----(i) T5=ar^4=5/2 -----(ii)
Now, dividing (ii) by dividing (i), we get r = ± 1/2 and putting it in (i) we get a = 40.
Now the required G.P is 40,20,10,5,5/2,... OR 40, -20, 10, -5, 5/2 , …...

(ii) Solution:
Since Tn = ar^(n-1)
T5=ar^4=8 ---- (i) T8 =ar^7= -64/27 -----(ii)
Now, dividing (ii) by dividing (i) we get r = - 2/3 and putting it in (i) we get a = 81/2.
Now the required G.P is 81/2 , -27,18, -12,8,...

(b)
(i) Solution:
Since Tn = a+(n-1)d ;
T4=a+3d=7……(i) T16=a+15d=31……(ii)
Now subtracting (ii) from (i), we get d = 2 and putting it in (i) we get a = 1.
Now the required A.P is 1,3,5,7,9, 11,...

(ii) Solution:
Since Tn = a+(n-1)d ;
T5=a+4d=86……(i) T10=a+9d=146……(ii)
Now subtracting (ii) from (i), we get d = 12 and putting it in (i) we get a = 38.
Now the required A.P is 38,50,62,74,86,…

c) Solution:
First, we find the A.P with the common difference (d)= 7
259,266,273, 280, ……. , 784
Since Tn = a+(n-1)d ; 784=259+(n-1)(7) ; n = 76.
Now for Sum; Sn = n/2 [2a+(n-1)d];
S_{76}= 76/2 [2(259) + (76-1)(7)] = 39,634.

d) Solution:

Since, $S_n = \frac{n}{2}[2a + (n-1)d]$ **------ (i)**
1st we have to find "d"
Now, $T_n = a + (n-1)d$

$$\frac{n^2 - n + 1}{n} = \frac{1}{n} + (n-1)d$$

Finally, d = 1. Hence putting it in we get,
$S_n = \frac{n^2-n+2}{2}$.

11. **a) Solution:**
The lower limit for the index of summation is 1, and the upper limit is 100. We write this sum as $\sum_{j=1}^{100} \frac{1}{j}$.

b)
(i) Solution: = $(-1)^4 + (-1)^5 + (-1)^6 + (-1)^7 + (-1)^8 = 1 + (-1) + 1 + (-1) + 1 = 1$.
(ii) Solution: = $1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1 + 4 + 9 + 16 + 25 = 55$.

c)
i) Solution:

$$a_0 = -1$$
$$a_1 = -2a_0 = -2(-1) = 2$$
$$a_2 = -2a_1 = -2(2) = -4$$
$$a_3 = -2a_2 = -2(-4) = 8$$
$$a_4 = -2a_3 = -2(8) = -16$$
$$a_5 = -2a_4 = -2(-16) = 32$$

ii) **Solution:**

$$a_0 = 2$$
$$a_1 = -1$$
$$a_2 = a_1 - a_0 = -1 - 2 = -3$$
$$a_3 = a_2 - a_1 = -3 - (-1) = -2$$
$$a_4 = a_3 - a_2 = -2 - (-3) = 1$$
$$a_5 = a_4 - a_3 = 1 - (-2) = 3$$

iii) **Solution:**

$$a_0 = 1$$
$$a_1 = 3a_0^2 = 3(1^2) = 3$$
$$a_2 = 3a_1^2 = 3(3^2) = 3(9) = 27$$
$$a_3 = 3a_2^2 = 3(27^2) = 3(729) = 2187$$
$$a_4 = 3a_3^2 = 3(2187^2) = 3(4782969) = 14348907$$
$$a_5 = 3a_4^2 = 3(14348907^2) = 3(205891132094649) = 617673396283947$$

iv) **Solution:**

$$a_0 = -1$$
$$a_1 = 0$$
$$a_2 = 2a_1 + a_0^2 = 2(0) + (-1)^2 = 0 + 1 = 1$$
$$a_3 = 3a_2 + a_1^2 = 3(1) + 0^2 = 3 + 0 = 3$$
$$a_4 = 4a_3 + a_2^2 = 4(3) + 1^2 = 12 + 1 = 13$$
$$a_5 = 5a_4 + a_3^2 = 5(13) + 3^2 = 65 + 9 = 74$$

d) **Solution:**

i) $a_0 = (-2)^0 = 1,\ a_1 = (-2)^1 = -2,\ a_2 = (-2)^2 = 4,\ a_3 = (-2)^3 = -8$

ii) $a_0 = a_1 = a_2 = a_3 = 3$

iii) $a_0 = 7 + 4^0 = 8,\ a_1 = 7 + 4^1 = 11,\ a_2 = 7 + 4^2 = 23,\ a_3 = 7 + 4^3 = 71$

iv) $a_0 = 2^0 + (-2)^0 = 2,\ a_1 = 2^1 + (-2)^1 = 0,\ a_2 = 2^2 + (-2)^2 = 8,\ a_3 = 2^3 + (-2)^3 = 0$

12. a) **Solution:**

Here a = 40, d = $T_2 - T_1$ = 33 – 40 = –7, n = 5, T_5 =?

$\boldsymbol{T_n = a + (n-1)d}$; $\boldsymbol{T_5 = 40 + (5-1)(-7) = 12}$.

The ball will fall 12 feet in 5th Second.

Now, $\boldsymbol{S_n = \frac{n}{2}[2a + (n-1)d]}$

$\boldsymbol{S_5 = \frac{5}{2}[2(40) + (5-1)(-7)]}$

$\boldsymbol{S_5 = 130}$. Hence ball covered 130 feet.

b) **Solution:**

Present population = a = 70000; Number of years = n = 6; Number of terms = n = 7

Rate of increase of population = r = 1 + 7% = 1.07

Population after 6 years = T_7 =?

$\boldsymbol{T_n = ar^{n-1} = 70000(1.07)^{7-1}}$ = 105051 approx.

The population after 7 years will be 105051.

13. i) **Solution:**

a) It has undirected edges.
It has multiple edges.
It has no loops.
It is undirected Multigraph.

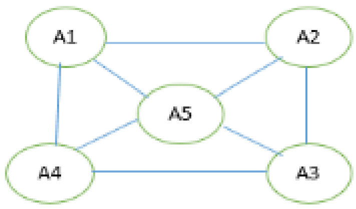
b) It has undirected edges.
It has no multiple edges.
It has no loops.
It is undirected simple graph.

c) It has undirected edges.
It has multiple edges.
It has three loops.
It is undirected Pseudo graph.

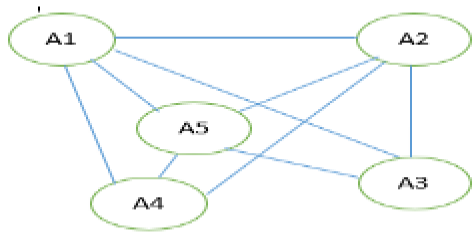
d) It has directed edges.
It has multiple edges.
It has two loops.
It is directed Multi graph.

ii) Solution:

a)



b)



14. a) Solution:

i) Number of Vertices: 5 Number of edges:13
Degree of vertices: deg(a) = deg(b) = deg(c) = 6, deg(d) = 5, deg(e) = 3.
Neighborhood Vertices: $N(a)$ = {a, b, e}, $N(b)$ = {a, c, d, e}, $N(c)$ = {b, c, d}, $N(d)$ = {b, c, e}, $N(e)$ = a, b, d}

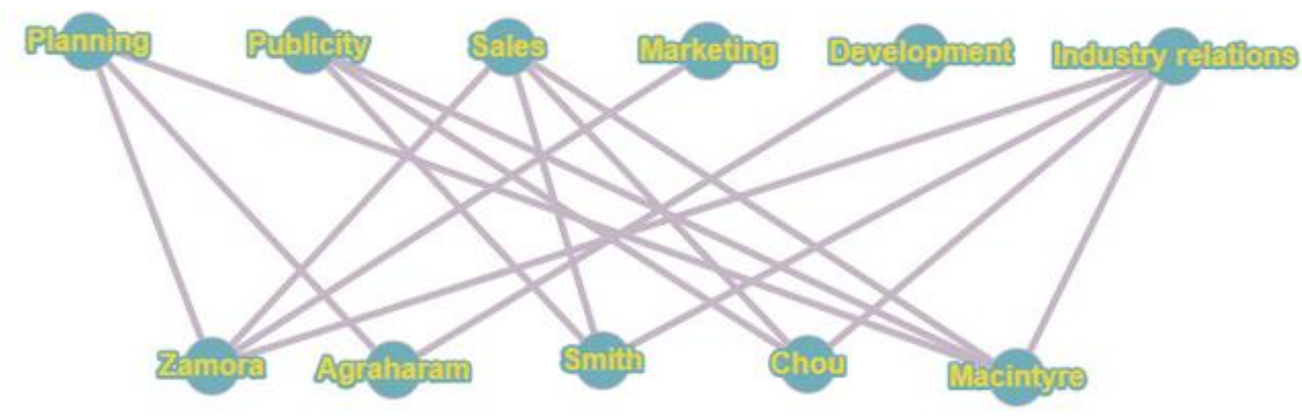
ii) Number of Vertices: 9 Number of edges:12
Degree of vertices: deg(a) =3, deg(b) =2, deg(c) = 4, deg(d) = 0, deg(e) = 6, deg(f) =0, deg(g) =4, deg(h) = 2, deg(i) = 3.
Neighborhood Vertices:
$N(a)$ = {c, e, i}, $N(b)$ = {e, h}, $N(c)$ = {a, e, g, i}, $N(d) = \varnothing$, $N(e)$ = {a, b, c, g}, $N(f) = \varnothing$, $N(g)$ = {c, e}, $N(h)$ = {b, i}, $N(i)$ = {a, c, h}.

b) Solution:
i) Number of Vertices: 5 Number of edges:13
In-degree of a vertices: $deg^-(a) = 6$, $deg^-(b) = 1$, $deg^-(c) = 2$, $deg^-(d) = 4$, $deg^-(e) = 0$.
Out-degree of a vertices: $deg^+(a) = 1$, $deg^+(b) = 5$, $deg^+(c) = 5$, $deg^+(d) = 2$, $deg^+(e) = 0$.

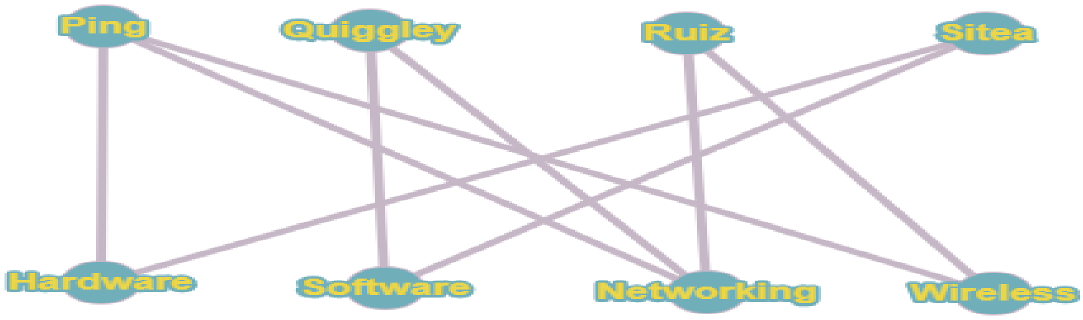
ii) Number of Vertices: 4 Number of edges:8
In-degree of a vertices: $deg^-(a) = 2$, $deg^-(b) = 3$, $deg^-(c) = 2$, $deg^-(d) = 1$.
Out-degree of a vertices: $deg^+(a) = 2$, $deg^+(b) = 4$, $deg^+(c) = 1$, $deg^+(d) = 1$.

c) Solution:
i) Only (W) and (Z) are connected, (Y) is disconnected; its connected components are {A, D, E} and {B, C}. (X) is disconnected; its connected components are {A, B, E} and {C, D}.
ii) Only (W) and (X) are cycle-free. (Y) has the cycle (A, D, E, A), and (Z) has the cycle (A, B, E, A).
iii) Only (X) has a loop which is {B, B}.
iv) Only (W) and (Y) are graphs. Multigraph (Z) has multiple edges {A, E} and {A, E}; and (X) has both multiple edges {C, D} and {C, D} and a loop {B, B}.

15. i) a) Solution: Bipartite Graph



b) Solution: Bipartite Graph



ii) Solution:

a) Not bipartite (since a is adjacent to b & f vertices)

b) Bipartite (A (V_1, V_3, V_5) & B (V_2, V_4, V_6))

c) Not bipartite (since V_4 & V_5 are adjacent vertices)

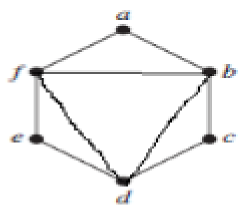
d) Not Bipartite (since b is adjacent to d & e vertices)

iii) Solution:

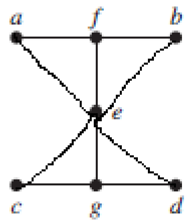
Graph G is Bipartite since two set of disjoint vertices exists. A{v, t, x} and B{u, w}
Graph H is Bipartite since two set of disjoint vertices exists. A{b, e, f} and B{a, c, d}
Graph A is not Bipartite since two set of disjoint vertices does not exists. Vertex y and z are adjacent.
Graph B is Bipartite since two set of disjoint vertices exists. A{d} and B{e, f, g, h, i, j, k, l}

16. a) Solution:

i)



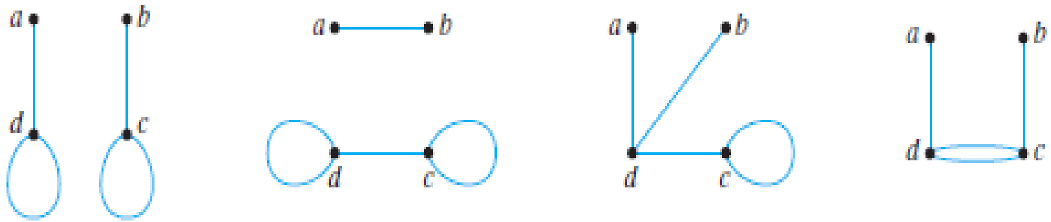
ii)



b) Solution:

i) No such graph is possible. By Handshaking theorem, the total degree of a graph is even.
But a graph with four vertices of degrees 1, 1, 2, and 3 would have a total degree of 1 + 1 + 2 + 3 = 7, which is odd.

ii) Let G be any of the graphs shown below.



In each case, no matter how the edges are labeled, deg(a) = 1, deg(b) = 1, deg(c) =3, and deg(d) = 3.

iii) There is no simple graph with four vertices of degrees 1, 1, 3, and 3.

c) Solution:

i) By using Handshaking theorem.
No! there is no graph possible, such that 15 vertices have degree 3. Since $(15 * 3) \neq 2e$.

ii) By using Handshaking theorem.
Yes! there is graph possible, such that 4 vertices have degree 3. Since $(4 * 3) = 2e$.

iii) This is asking for the number of edges in K10. Each vertex (person) has degree (shook hands with) 9 (people). So the sum of the degrees is 90. However, the degrees count each edge (handshake) twice, so there are 45 edges in the graph. That is how many handshakes took place.

iv) It is possible for everyone to be friends with exactly 2 people. You could arrange the 5 people in a circle and say that everyone is friends with the two people on either side of them (so you get the graph C5). However, it is not possible for everyone to be friends with 3 people. That would lead to a graph with an odd number of odd degree vertices which is impossible since the sum of the degrees must be even.

v)

We want to determine a regular graph of degree four with $m = 10$ edges.

Let the graph contain n vertices $v_1, v_2, ..., v_n$, then each of these n vertices have degree 4.

$$\deg(v_i) = 4$$
$$i = 1, 2, ..., n$$

By the Handshaking theorem, the sum of degrees of all vertices is equal to twice the number of edges:

$$20 = 2(10) = 2m = \sum_{v=1}^{n} \deg(v_i) = \sum_{v=1}^{n} 4 = 4n$$

We then obtained the equation $20 = 4n$. Divide each side of the equation by 4:

$$n = \frac{20}{4} = 5$$

17. i)

a) Solution:

Both graph G and G' are satisfying all the invariant. Hence, they are isomorphic.

Function: $g(V_1) = W_2$, $g(V_2) = W_3$ $g(V_3) = W_1$ $g(V_4) = W_5$ $g(V_5) = W_4$

b) Solution: Both graph G and G' are satisfying all the invariant. Hence, they are isomorphic.

Function: $g(V_1) = U_5$, $g(V_2) = U_2$ $g(V_3) = U_4$ $g(V_4) = U_3$ $g(V_5) = U_1$ $g(V_6) = U_6$

c) Solution: Both graph G and G' are satisfying all the invariant. Hence, they are isomorphic.

Function: $g(V_1) = U_5$, $g(V_2) = U_4$ $g(V_3) = U_3$ $g(V_4) = U_2$
$g(V_5) = U_7$ $g(V_6) = U_1$ $g(V_7) = U_6$

d) Solution: Graph G has no vertex of degree 4 where G' has vertex V_2 with degree 4. Hence, they are not isomorphic.

ii) Solution:

We note that A and R are isomorphic graphs. Also, F and T are isomorphic graphs, K and X are isomorphic graphs and M, S, V, and Z are isomorphic graphs.

18. a) Solution:

N	D(b)	D (c)	D(d)	D(e)	D(f)	D(g)	D(h)	D(i)	D(j)	D(k)	D(l)	D(m)	D(n)	D(o)	D(p)	D(q)	D(r)	D(s)	D(t)	D(z)
a	2,a	4,a	1,a																	
ad	2,a	4,a			6,d	5,d														
adb		4,a		3,b	6,d	5,d														
adbe		4,a			6,d	5,d	6,e													
adbec					6,c	5,d	6,e													
adbeg					6,c		6,e			7,g										
adbegf							6,e	8,f	10,f	7,g										
adbegfh								8,f	10,f	7,g	7,h			14,h						
adbegfhk								8,f	10,f		7,h		11,k	14,h			9,k			
adbegfhkl								8,f	10,f			10,l	11,k	13,l			9,k			
adbegfhkli									10,f			10,l	11,k	13,l			9,k			
adbegfhklir									10,f			10,l	10,r	13,l		17,r			14,r	
adbegfhklirj												10,l	10,r	13,l		17,r			14,r	
adbegfhklirjm													10,r	13,l	12,m	17,r			14,r	
adbegfhklirjmn														13,l	12,m	12,n			14,r	
adbegfhklirjmnp														13,l		12,n		14,p	13,p	
adbegfhklirjmnpq														13,l				14,p	13,p	
adbegfhklirjmnpqo																		14,p	13,p	
adbegfhklirjmnpqot																		14,p		21,t
adbegfhklirjmnpqots																				16,s
adbegfhklirjmnpqotsz																				

b) Solution:

N	D(b)	D(c)	D(d)	D(e)	D(f)	D(g)	D(z)
a	4,a	3,a	∞	∞	∞	∞	∞
ac			6,c	9,c	∞	∞	∞
acb			6,c	9,c	∞	∞	∞
acbd				7,d	11,d	∞	∞
acbde					11,d	12,e	∞
acbdef						12,e	18,f
acbdefg							16,g
acbdefgz	4,a	3,a	6,c	7,d	11,d	12,e	16,g

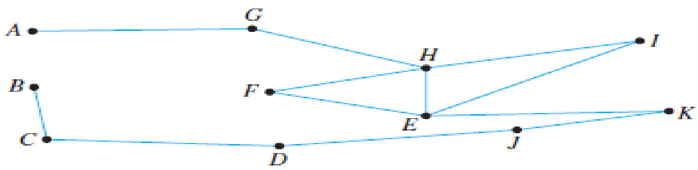
c) Solution:

N	D(b)	D(c)	D(d)	D(e)	D(f)	D(z)
a	12,a	10,a	∞	∞	12,a	∞
ac	12,a		21,c	13,c	12,a	∞
acb			21,c	13,c	12,a	∞
acbf			21,c	13,c		21,f
acbfe			21,c			19,e
acbfez			21,c			
acbfezd	12,a	10,a	21,c	13,c	12,a	19,f

19. a) Solution: Yes! Path: A→H→G→B→C→D→G→F→E

b)

Solution Let the floor plan of the house be represented by the graph below.



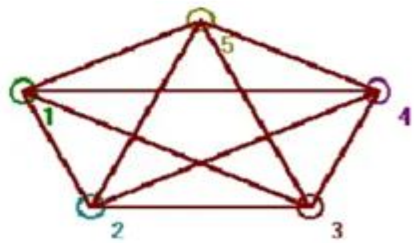
Each vertex of this graph has even degree except for A and B, each of which has degree 1. Hence by Corollary 10.2.5, there is an Euler path from A to B. One such trail is

$AGHFEIHEKJDCB$.

c) Solution:

This is s a Euler graph as 1-2-3-1-5-2-4-3-5-4-1

{(1,2), (2,3), (1,3), (1,5), (2,5), (2,4), (3,4), (3,5), (4,5), (1,4)}



20. a) Solution:

i) Hamiltonian Circuit: $V_0, V_1, V_2, V_6, V_5, V_4, V_7, V_3, V_0$. Hamiltonian Path: $V_0, V_1, V_2, V_6, V_5, V_4, V_7, V_3$

ii) Hamiltonian Circuit: doesn't exist. Hamiltonian Path: b, c, f, g, h, e, a, d

iii) Hamiltonian Circuit: d, c, b, a, g, f, e, d. Hamiltonian Path: d, c, b, a, g, f, e

b) Solution:

i) Hamiltonian Circuit: a, b, c, d, e, f, g, h, a. Hamiltonian Path: a, b, c, d, e, f, g, h.

ii) Hamiltonian Circuit: doesn't exist. Hamiltonian Path: doesn't exists

iii) Hamiltonian Circuit: a, x, b, y, c, z, a. Hamiltonian Path: a, x, b, y, c, z.

c) Solution:

i) Hamiltonian Circuit are: ABCDA = 125; ABDCA = 140; ACBDA = 155.

Hence ABCDA = 125 is the minimum distance travelled.

ii) Hamiltonian Circuit are: ABCDA = 97; ABDCA = 108; ACBDA = 141.

Hence ABCDA = 97 is the minimum distance travelled.

iii) Hamiltonian Circuit are: ABCDA = 95; ABDCA = 80; ACBDA = 95.

Hence ABDCA = 80 is the minimum distance travelled.

21. a) Solution:

i) All vertices have even degree so circuit exists. Euler Circuit: V_1, V_2, V_5, V_4, V_5, V_2, V_3, V_4, V_1

ii) Euler Circuit do not exist because all vertices don't have even degree. Vertex V_9 and V_7 has odd degree.

iii) All vertices have even degree so circuit exists. Euler Circuit: a, b, c, a, i, b, h, d, c, i, h, g, d, e, f, g, c, e, h, a.

b) Solution:

i) Euler Path doesn't exist because four vertices have odd degree.

ii) Euler Path exists because exact two vertices have odd degree.

Euler path: U, V_1, V_0, V_7, U, V_2, V_3, V_4, V_2, V_6, V_5, W, V_6, V_4, W.

22. a)

Solution: i)

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

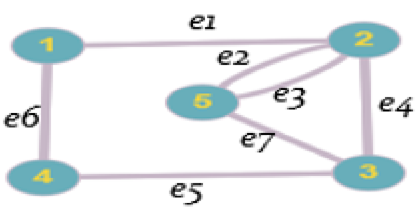
ii)

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

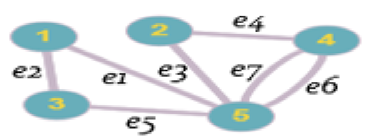
b)

Solution:

i)



ii)



c) Solution:

i)

Initial Vertex	Terminal Vertices
a	a, b, c, d
b	d
c	a, b
d	b, c, d

ii)

Initial Vertex	Terminal Vertices
a	b, d
b	a, c, d, e
c	b, c,
d	a, e
e	c, e

iii)

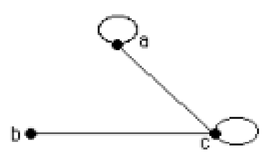
Vertex	Adjacent Vertices
a	b, d
b	a, c
c	b, d
d	a, c

iv)

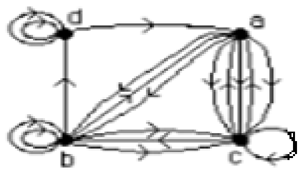
Vertex	Adjacent Vertices
a	a, c, d
b	b, c, d
c	a, b, c
d	a, c, d

d) Solution:

i)

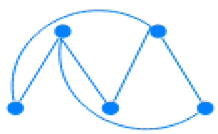


ii)



23. a) Solution:

i) This graph can be untangled if we play with it long enough. The following picture gives a planar representation of it.



ii) This graph is not planar because we can form a $K_{3,3}$ graph with the vertices {a, d, f} and {b, c, e}.

b) Solution: Graph G: V = 4, E = 6, R = 4. Hence, R = E − V + 2 = 6 − 4 + 2 = 4.
Solution: Graph H: V = 6, E = 9, R = 5; Hence, R = E − V + 2 = 9 − 6 + 2 = 5.
Solution: Graph J: V = 5, E = 10, R = 7. Hence, R = E − V + 2 = 10 − 5 + 2 = 7.

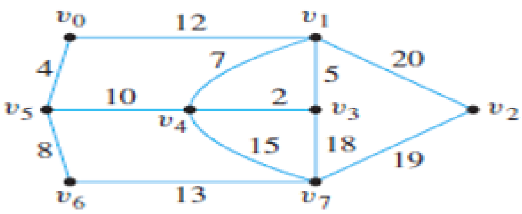
24. i) Solution:

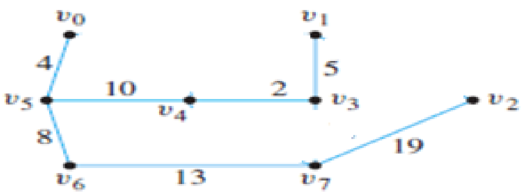
a) Level of n is 3. b) Level of a is 0. c) Height of this rooted tree is 5
d) u & v are the children of n. e) d is the parent of g. f) k & l are the siblings of j.
g) m, s, t, x & y are the descendants of f. h) a, b, e, k, c, f, m, t, d, h, i, n, o & v are the internal nodes.
i) v, n, h, d & a are the ancestors of z. j) j, l, q, r, s, x, y, g, p, u, w & z are the leaves.

ii) Solution:
a) A tree with n vertices has n – 1 edge. Hence 10000 – 1 = 9999 edges.
b) A full binary tree has two edges for each internal vertex. So, we'll just multiply the number of internal vertices by the number of edges. Hence 1000 * 2 = 2000 edges.
c) A full m-ary tree with i internal vertices has n = mi + 1 vertices. Since, we have m = 5, i = 100.
Hence n = 5*100 + 1 = 501 vertices.
d) A full m-ary tree with n vertices has $l = [(m - 1)(n +)1]/m$ *leaves*. Since, we have m = 3, n = 100.
Hence $l = [(3 - 1)(100 + 1)] / 2$ = 101 leaves.

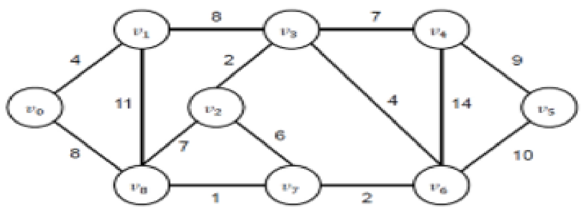
25. a) Solution:

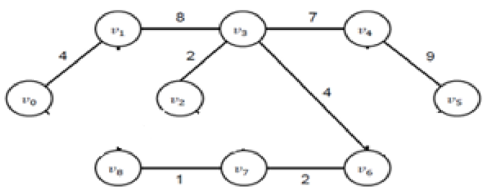
i) Solution: MST Cost = 61
$(V_0,V_5) = 4$, $(V_5,V_6) = 8$, $(V_4,V_5) = 10$,
$(V_3,V_4) = 2$, $(V_1,V_3) = 5$, $(V_6,V_7) = 13$,
$(V_2,V_7) = 19$.

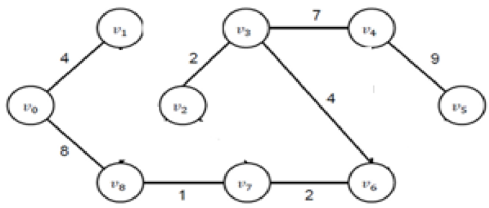




ii) Solution: MST Cost = 37
$(V_0,V_1) = 4$, $(V_0,V_8) = 8$, $(V_7,V_8) = 1$,
$(V_6,V_7) = 2$, $(V_3,V_6) = 4$, $(V_2,V_3) = 2$,
$(V_3,V_4) = 7$, $(V_4,V_5) = 9$,

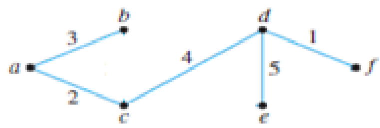




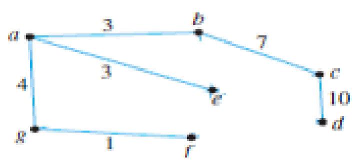


b)

i) Solution: MST cost = 15
Order of edges added is:
(d, f) =1, (a, c) =2, (a, b) =3, ~~(b, c) =3,~~
(c, d) =4, (d, e) =5, ~~(c, e) =6,~~ ~~(e, f) =6~~



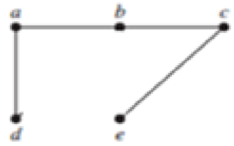
ii) Solution: MST cost = 28
Order of edges added is:
(g, f) =1, (a, b) =3, (a, e) =3, (a, g) =4,
~~(b, e) =4,~~ ~~(e, f) =4,~~ ~~(g, e) =5,~~ (b, c) =7
~~(c, e) =7,~~ (c, d) =10, ~~(d, e) =11,~~ ~~(d, f) =12~~



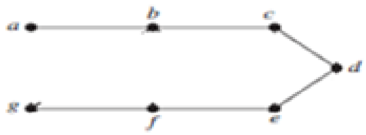
c)

Solution:

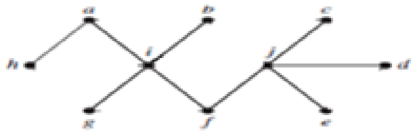
i)



ii)

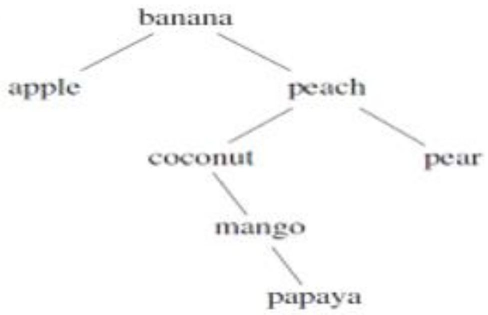


iii)

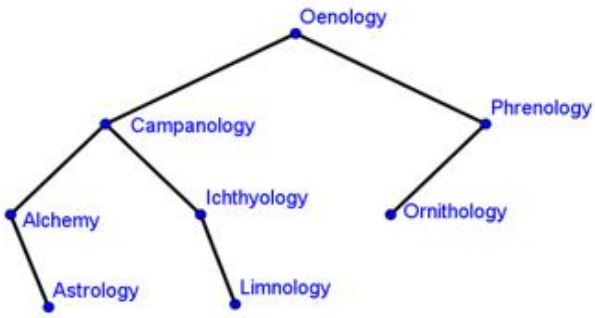


26. a) **Solution:**

i)

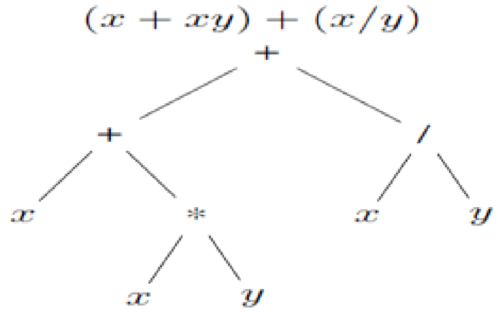


ii)

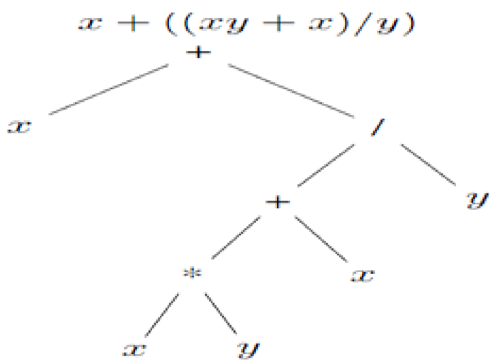


b) **Solution:**

i)



ii)



27. a) **Solution:**

i)

Pre order: a b e k l m f g n r s c d h o l j p q
In order: k e l m b f r n s g a c o h d i p j q
Post order: k l m e f r s n g b c o h l p q j d a

ii)

Pre order: a b d e i j m n o c f g h k l p
In order: d b i e m j n o a f c g k h p l
Post order: d i m n o j e b f g k p l h c a

b) **Solution:**

i) Prefix: + + x * x y / x y   Postfix: x x y * + x y / +

ii) Prefix: + x / + * x y x y   Postfix: x x y * x + y / +

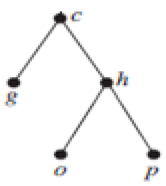
c) **Solution:** i) 4   ii) 3

d) **Solution:**

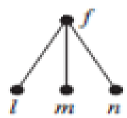
i) It is not a full m-ary tree for any m because some of its internal vertices have two children and others have three children.

ii) It is not balanced m-ary tree because it has leaves at levels 2, 3, 4 and 5.

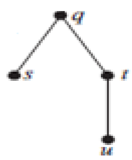
iii) a)



b)

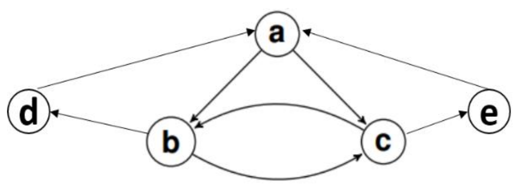


c)



28. **Solution:**

a)



b) In Degree: $\deg^-(a) = 2$, $\deg^-(b) = 2$, $\deg^-(c) = 2$, $\deg^-(d) = 1$, $\deg^-(e) = 1$
Out Degree: $\deg^+(a) = 2$, $\deg^+(b) = 2$, $\deg^+(c) = 2$, $\deg^+(d) = 1$, $\deg^+(e) = 1$

c) For equivalence relation, relation should be reflexive, Symmetric and Transitive.
For partial Order relation, relation should be reflexive, Antisymmetric and Transitive.

Relation	Reason
Not reflexive	Because every vertex does not have a self-loop.
Not symmetric	Because we do not have (b, a) for (a, b).
Not antisymmetric	Because we have (b, c) and (c, b).
Not transitive	Because the path for (b, c), (c, b) from b to b is missing the edge (b, b).

Hence, the relation is neither Equivalence nor Partial order relation.

29. **Solution:**

a) The shortest path, which could be found using Dijkstra's algorithm, is Home →B→D→F→ School
Cost: 2 + 1 + 4 + 2 = 9.

N	*D(A)*	*D(B)*	*D(C)*	*D(D)*	*D(E)*	*D(F)*	*D(S)*
H	*3,H*	*2,H*	5,H	∞	∞	∞	∞
HB	*3,H*		5,H	*3,B*	*8,B*	∞	∞
HBA			5,H	*3,B*	*8,B*	∞	∞
HBAD			5,H		*8,B*	7,D	∞
HBADC					*7,C*	7,D	∞
HBADCE						7,D	*11,E*
HBADCEF							*9,F*
HBADCEFS							

b) No Euler Circuit possible because some nodes have odd vertices. No Euler path is possible because there are more than two vertices with odd degree.

c) No Hamilton Circuit But path is possible. E.g: C-> E-> S -> F-> D-> B-> H-> A

30. **Solution:**

a) Both Graph G and H are isomorphic. f(A) = 1, f(B) = 6, f(C) = 8, f(D) = 3, f(W) = 5, f(X) = 2, f(Y) = 4, f(Z) = 7

b)

i) Graph G is not complete bipartite as A is not adjacent with Z, B is not adjacent with Y, C is not adjacent with X, and D is not adjacent with W.

ii) Graph H doesn't contain Euler Circuit as all the vertices have odd degree.

Graph H Contains Hamilton's Circuits:

H1: 1,2,3,7,6,5,8,4,1 H2: 1,2,6,7,3,4,8,5,1 H3: 1,2,6,5,8,7,3,4,1